

Mahir Dalal

Surat, India | mahirdalal97@gmail.com | +91 7624023777

 [GitHub](#) |  [LinkedIn](#) |  [Portfolio](#)

Summary

Computer Science undergraduate at IIIT Surat with strong foundations in competitive programming, software engineering, full-stack development, and AI-integrated applications. Experienced in building scalable web platforms, distributed systems, agentic AI systems, concurrent backend architectures and machine learning-integrated systems.

Education

Indian Institute of Information Technology, Surat

2024 – 2028

B.Tech in Computer Science and Engineering

CGPA: **8.05 / 10** | JEE Main Percentile: **98.2%**

Technical Skills

Languages: C++, Java, Python, JavaScript, TypeScript

Web Development: Next.js, React, Tailwind CSS, Node.js, Express.js, NestJS, FastAPI, REST APIs

Databases & Tools: MongoDB, PostgreSQL, SQL, Git, GitHub, Docker, Postman

AI & Agentic Systems: LangChain, LangGraph, LLM Integration, AI Workflow Orchestration

Systems & Architecture: Socket.IO, WebSockets, Event-Driven Architecture, Real-Time Systems, Distributed Systems, Concurrency, Multithreading, Synchronization, Performance Optimization

Computer Science Fundamentals: Data Structures & Algorithms, Database Management Systems, Operating Systems, Computer Networks, OOP

Projects

AI DSA/CP Coach

Next.js, FastAPI, PostgreSQL, LangGraph, LangChain

- Built an AI-powered adaptive learning platform for DSA and Competitive Programming that generates personalized practice plans using learner behavior, topic mastery, and contest-performance signals.
- Designed a hybrid recommendation engine leveraging **10+ learning signals** including solve history, solve time, rating progression, hint dependency, and recurring mistake trends, reducing manual effort required for problem selection and study planning by an estimated **65%**.
- Built multi-agent coaching workflows using LangGraph and LangChain for contextual hint generation, failure analysis, revision planning, and automated learning-path orchestration, enabling personalized learning recommendations at scale.
- Engineered a structured learner-modeling and analytics system tracking **20+ performance metrics** across DSA and Competitive Programming workflows, reducing time spent identifying learning weaknesses by an estimated **50%** through automated feedback and adaptive progression tracking.

JAN-AI

Next.js, NestJS, PostgreSQL, Prisma, Mapbox, Groq API, TypeScript

- Built an AI-assisted civic infrastructure platform supporting **5+ infrastructure categories** including roads, sanitation, drainage, streetlights, and public spaces through geotagged image-based reporting and location-aware issue tracking.
- Engineered a scalable, production-grade full-stack architecture using Next.js, NestJS, PostgreSQL, and Prisma, processing **10+ structured metadata attributes** per report for infrastructure monitoring and analytics.
- Integrated multimodal AI services for automated issue categorization, severity assessment, report summarization, and content moderation, reducing manual report analysis effort by an estimated **60%**.

Intelligent Self-Healing Global Network Simulator

Node.js, Express, React 19, Three.js, Socket.IO

- Architected a distributed real-time network simulation across **15 global nodes and 21 edges** using a 6-service event-driven backend with multi-threaded request handling, broadcasting synchronized full state to concurrent clients via WebSocket every 300ms.
- Implemented dual concurrent routing algorithms (naive OSPF-style vs. health-aware Dijkstra) running in parallel with autonomous mid-transit rerouting when node failure probability exceeds threshold.
- Engineered a reinforcement-learning-inspired node scoring system with dynamic health feedback loops across **4 attack models** (Blackhole, Grayhole, Delay, DDoS Flood), enabling the routing layer to autonomously exclude compromised nodes without manual intervention.
- Designed a multi-priority packet scheduling system with configurable cost weights and a composite routing cost function balancing latency, load, and node trust score across a live concurrent simulation.

Competitive Programming

- Codeforces:** Specialist (Max: 1403)
- LeetCode:** Knight (Max Rating: 1879)
- CodeChef:** 3 Star (Max: 1618)
- AtCoder:** 7-kyu (Max: 601)

[CF Profile](#) ↗

[LC Profile](#) ↗

[CC Profile](#) ↗

[AC Profile](#) ↗

Experience

Competitive Programming Mentor, Learn Code Solve (LCS)

Aug 2025 – Present

- Collaborated with competitive programmers to design, review, and validate algorithmic problems for contests and practice sets.
- Contributed to contest organization, editorials, solution verification, and problem-setting initiatives.

Achievements

- Ranked in the Top **5%** globally on LeetCode (Peak Contest Rating: 1879, Knight).
- Solved **1000+** Data Structures & Algorithms problems across multiple online judges.
- Achieved Global Rank 134 in CodeChef Starters 218.
- Ranked in the **Top 2%** of 1.4M+ candidates in JEE Main (98.2 percentile).